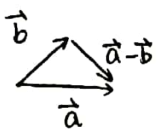
8.3 向量应用（一）平面及方程

回顾：一、向量运算的几何描述



1. $\vec{a}+\vec{b}$:
2. $\vec{a}-\vec{b}$:
3. $k\vec{a}\begin{cases} k>0 \\ k<0 \\ k=0 \end{cases}$
4. 向量的数量积
$\vec{a}\cdot\vec{b} \triangleq |\vec{a}||\vec{b}|\cos\langle\vec{a},\vec{b}\rangle$.
5. $\vec{a}\times\vec{b}\begin{cases} \text{方向: 右手准则} \\ \text{大小: } |\vec{a}\times\vec{b}|=|\vec{a}||\vec{b}|\sin\langle\vec{a},\vec{b}\rangle \end{cases}$

二、向量运算的代数描述

设 $\vec{a}=\{a_1,b_1,c_1\}$ $\vec{b}=\{a_2,b_2,c_2\}$

1. $\vec{a}+\vec{b}=\{a_1+a_2,\ b_1+b_2,\ c_1+c_2\}$
2. $\vec{a}-\vec{b}=\{a_1-a_2,\ b_1-b_2,\ c_1-c_2\}$.
3. $k\vec{a}=\{ka_1,\ kb_1,\ kc_1\}$.
4. $\vec{a}\cdot\vec{b}=a_1a_2+b_1b_2+c_1c_2$
5. $\vec{a}\times\vec{b}=\{b_1c_2-b_2c_1,\ a_2c_1-a_1c_2,\ a_1b_2-a_2b_1\}$

$$\begin{matrix} a_1 & b_1 & c_1 & a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 & a_2 & b_2 & c_2 \end{matrix}$$

一、空间曲面

设 $F(x,y,z)=0$ 为一个三元方程

Σ 为一个曲面，若 $F(x,y,z)=0$

任一解 (x_0,y_0,z_0) 对应的点 $M_0(x_0,y_0,z_0)$ 在曲面 Σ 上；

反之，曲面 Σ 上任一点 $M(x_0,y_0,z_0)\Rightarrow$

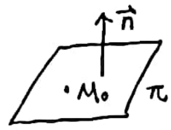
$F(x_0,y_0,z_0)=0$

称 $F(x,y,z)=0$ 为曲面 Σ 的方程，Σ 为方程 $F(x,y,z)=0$ 对应的曲面.

记为 $\Sigma: F(x,y,z)=0$

二、曲面的特殊情形——平面.

（一）平面的点法式方程



设 $M_0(x_0,y_0,z_0)\in\pi$

$\vec{n}=\{A,B,C\}\perp\pi$

$\forall M(x,y,z)$

$M(x,y,z)\in\pi \Leftrightarrow \vec{n}\perp\overrightarrow{M_0M} \Leftrightarrow \vec{n}\cdot\overrightarrow{M_0M}=0$

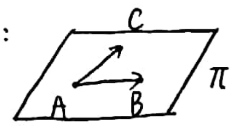
而 $\overrightarrow{M_0M}=\{x-x_0,\ y-y_0,\ z-z_0\}$

$\therefore \pi: A(x-x_0)+B(y-y_0)+C(z-z_0)=0$

例1：$A(-1,0,1)$ $B(0,1,2)$ $C(2,1,1)$

求：A、B、C 确定的平面 π.

解：



$\overrightarrow{AB}=\{1,1,1\}$

$\overrightarrow{AC}=\{3,1,0\}$

$\vec{n}=\overrightarrow{AB}\times\overrightarrow{AC}=\{-1,3,-2\}$

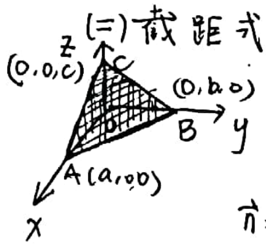
$$\begin{matrix} \cancel{1} & 1 & 1 & 1 & 1 & 1 \\ \cancel{3} & 1 & 0 & 3 & 1 & 0 \end{matrix}$$

$\therefore \pi: -1\times(x+1)+3\times(y-0)-2\times(z-1)=0$

即 π: $x+1-3y+2(z-1)=0$

$x-3y+2z-1=0$

（二）截距式方程



$\overrightarrow{AB}=\{-a,b,0\}$

$\overrightarrow{AC}=\{-a,0,c\}$

$\vec{n}=\overrightarrow{AB}\times\overrightarrow{AC}=\{bc,ac,ab\}$

π: $bc(x-a)+ac(y-0)+ab(z-0)=0$

即 π: $\frac{x-a}{a}+\frac{y}{b}+\frac{z}{c}=0$

即 $\pi: \frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$

(三) 一般式方程

$Ax+By+Cz+D=0$

$Ax_0+By_0+Cz_0+D=0$

$\Rightarrow A(x-x_0)+B(y-y_0)+C(z-z_0)=0$

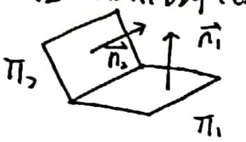
$\pi: Ax+By+Cz+D=0$ (一般式方程)

且 $\vec{n}=\{A,B,C\}$

三. 两个平面的夹角.

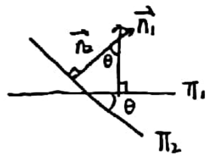
$\pi_1: A_1x+B_1y+C_1z+D_1=0$

$\pi_2: A_2x+B_2y+C_2z+D_2=0$



$\vec{n_1}=\{A_1,B_1,C_1\}$

$\vec{n_2}=\{A_2,B_2,C_2\}$



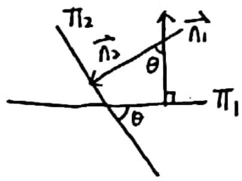
① $\langle\vec{n_1},\vec{n_2}\rangle\in[0,\frac{\pi}{2}]$

则 $\theta=\langle\vec{n_1},\vec{n_2}\rangle$

$\cos\theta=\cos\langle\vec{n_1},\vec{n_2}\rangle$

$=\frac{\vec{n_1}\cdot\vec{n_2}}{|\vec{n_1}||\vec{n_2}|}$

② $\langle\vec{n_1},\vec{n_2}\rangle\in[\frac{\pi}{2},\pi]$



则 $\theta+\langle\vec{n_1},\vec{n_2}\rangle=\pi \Rightarrow \theta=\pi-\langle\vec{n_1},\vec{n_2}\rangle$

$\cos\theta=-\cos\langle\vec{n_1},\vec{n_2}\rangle=-\frac{\vec{n_1}\cdot\vec{n_2}}{|\vec{n_1}||\vec{n_2}|}$

统一形式：$\cos\theta=\left|\frac{\vec{n_1}\cdot\vec{n_2}}{|\vec{n_1}||\vec{n_2}|}\right|$

例3：$\pi_1: x-y+2z-6=0$ 求夹角.
$\pi_2: 2x+y+z-5=0$

解：设夹角为 θ $(0\le\theta\le\frac{\pi}{2})$

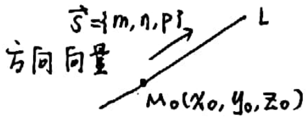
$\vec{n_1}=\{1,-1,2\}$ $\vec{n_2}=\{2,1,1\}$

$\cos\theta=\left|\frac{\vec{n_1}\cdot\vec{n_2}}{|\vec{n_1}||\vec{n_2}|}\right|=\frac{1}{2}$

$\Rightarrow\theta=\frac{\pi}{3}$

8.4. 向量应用(二) — 空间直线

一. 点向式(对称式)方程



$M_0(x_0,y_0,z_0)\in L$

$\vec{s}=\{m,n,p\}\parallel L$

$\forall M(x,y,z)$ $M\in L\Leftrightarrow\overrightarrow{M_0M}\parallel\vec{s}$

$\overrightarrow{M_0M}=\{x-x_0,y-y_0,z-z_0\}$

$\overrightarrow{M_0M}\parallel\vec{s}\Leftrightarrow\frac{x-x_0}{m}=\frac{y-y_0}{n}=\frac{z-z_0}{p}$

$\therefore L: \frac{x-x_0}{m}=\frac{y-y_0}{n}=\frac{z-z_0}{p}$ (点向式)

例1. 设直线L经过点 ~~A(~~ $A(1,-1,0)$ $B(2,1,1)$ 求L的方程.

解：$\vec{s}=\overrightarrow{AB}=\{1,2,1\}$

$L: \frac{x-1}{1}=\frac{y+1}{2}=\frac{z-0}{1}$

二. 参数式方程.

设 $M_0(x_0,y_0,z_0)\in L$ $\vec{s}=\{m,n,p\}\parallel L$

$L: \frac{x-x_0}{m}=\frac{y-y_0}{n}=\frac{z-z_0}{p}$

令 $\frac{x-x_0}{m}=\frac{y-y_0}{n}=\frac{z-z_0}{p}=t$

∴ ~~L的参数式方程~~ $\cancel{x=x_0+mt}$ $\cancel{y=y_0+nt}$ $\cancel{z=}$

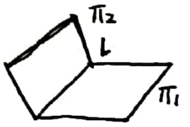
L的参数式方程

$$L:\begin{cases} x=x_0+mt \\ y=y_0+nt \\ z=z_0+pt \end{cases}$$

如：$L: \frac{x-1}{1}=\frac{y+1}{2}=\frac{z}{1}$ 参数式方程为

$$L:\begin{cases} x=1+t \\ y=-1+2t \\ z=t \end{cases}$$

三. 直线方程的一般式



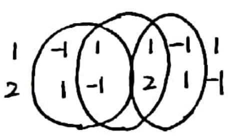
$$L:\begin{cases} A_1x+B_1y+C_1z+D_1=0 \\ A_2x+B_2y+C_2z+D_2=0 \end{cases}$$

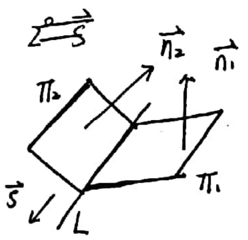
直线方程的一般式方程.

例2：设 $L:\begin{cases} x-y+z-2=0 \\ 2x+y-z-1=0 \end{cases}$ 化为点向式

解：1° $M_0(1,1,2)\in L$

2° $\vec{s}=\vec{n_1}\times\vec{n_2}$





$\vec{n_1}\perp L$ $\vec{n_2}\perp L$

$\vec{s}=\{0,3,3\}$

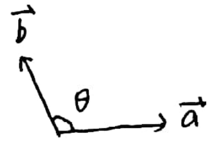
3° $L: \frac{x-1}{0}=\frac{y-1}{3}=\frac{z-2}{3}$

或 $L: \frac{x-1}{0}=\frac{y-1}{1}=\frac{z-2}{1}$

四. 杂知识点

(一) 夹角

1. 两个向量的夹角



设 $\vec{a},\vec{b}$ 为两个向量 $\langle\vec{a},\vec{b}\rangle=\theta\ (0\le\theta\le\pi)$

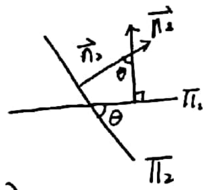
① 几何：由 $\vec{a}\cdot\vec{b}=|\vec{a}||\vec{b}|\cos\theta$.

得 $\cos\theta=\frac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|}$

② 代数：设 $\vec{a}=\{a_1,b_1,c_1\}$ $\vec{b}=\{a_2,b_2,c_2\}$.

$$\cos\theta=\frac{a_1a_2+b_1b_2+c_1c_2}{\sqrt{a_1^2+b_1^2+c_1^2}\cdot\sqrt{a_2^2+b_2^2+c_2^2}}$$

2. 两平面夹角.



$\pi_1: A_1x+B_1y+C_1z+D_1=0$

$\pi_2: A_2x+B_2y+C_2z+D_2=0$

设 π_1,π_2 的夹角为 θ (为锐角)

① $\langle\vec{n_1},\vec{n_2}\rangle\in[0,\frac{\pi}{2}]$ 时 $\theta=\langle\vec{n_1},\vec{n_2}\rangle$

② $\langle\vec{n_1},\vec{n_2}\rangle\in(\frac{\pi}{2},\pi)$ 时 $\theta+\langle\vec{n_1},\vec{n_2}\rangle=\pi$

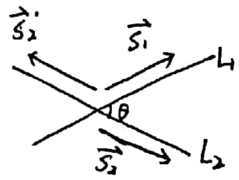
$\theta=\pi-\langle\vec{n_1},\vec{n_2}\rangle$

$\Rightarrow \cos\theta=|\cos\langle\vec{n_1},\vec{n_2}\rangle|=\frac{|\vec{n_1}\cdot\vec{n_2}|}{|\vec{n_1}||\vec{n_2}|}$

3. 两直线夹角

$L_1: \frac{x-x_1}{m_1}=\frac{y-y_1}{n_1}=\frac{z-z_1}{p_1}$

$L_2: \frac{x-x_2}{m_2}=\frac{y-y_2}{n_2}=\frac{z-z_2}{p_2}$.



不一定相交

设 L_1,L_2 夹角为 θ $(0\le\theta\le\frac{\pi}{2})$

① 若 $\langle\vec{s_1},\vec{s_2}\rangle\in[0,\frac{\pi}{2}]$

则 $\theta=\langle\vec{s_1},\vec{s_2}\rangle$

② 如果 $\langle\vec{s_1},\vec{s_2}\rangle\in(\frac{\pi}{2},\pi)$

则 $\theta=\pi-\langle\vec{s_1},\vec{s_2}\rangle$

$\cos\theta = |\cos\langle \vec{s_1},\vec{s_2}\rangle| = \frac{|\vec{s_1}\cdot\vec{s_2}|}{|\vec{s_1}||\vec{s_2}|}$

$$\cos\theta = \frac{|m_1m_2+n_1n_2+p_1p_2|}{\sqrt{m_1^2+n_1^2+p_1^2}\cdot\sqrt{m_2^2+n_2^2+p_2^2}}$$

例3: 求 $L_1: \frac{x-1}{1}=\frac{y}{-4}=\frac{z+3}{1}$

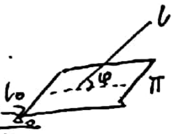
$L_2: \frac{x}{2}=\frac{y+2}{-2}=\frac{z}{-1}$ 夹角

解: $\vec{s_1}=\{1,-4,1\}$ $\vec{s_2}=\{2,-2,-1\}$

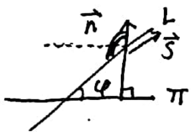
设 L_1、L_2 夹角为 θ, 则 $\cos\theta=|\cos\langle\vec{s_1},\vec{s_2}\rangle|$

$\cos\theta = \frac{|2+8-1|}{\sqrt{18}\sqrt{9}} = \frac{\sqrt{2}}{2}$ $\theta=\frac{\pi}{4}$

4. 直线与平面夹角



设 $L: \frac{x-x_0}{m}=\frac{y-y_0}{n}=\frac{z-z_0}{p}$



$\vec{s}=\{m,n,p\} \parallel L$

$\pi: Ax+By+Cz+D=0$

$\vec{n}=\{A,B,C\}$, 设 L 与 π 夹角为 φ $(0\le\varphi\le\frac{\pi}{2})$

① $\langle\vec n,\vec s\rangle\in[0,\frac{\pi}{2}]$ 时

$\varphi+\langle\vec n,\vec s\rangle=\frac{\pi}{2} \Rightarrow \varphi=\frac{\pi}{2}-\langle\vec n,\vec s\rangle$

$\sin\varphi=\cos\langle\vec n,\vec s\rangle$

② $\langle\vec n,\vec s\rangle\in(\frac{\pi}{2},\pi)$ 时

$\langle\vec n,\vec s\rangle=\frac{\pi}{2}+\varphi \Rightarrow \varphi=\langle\vec n,\vec s\rangle-\frac{\pi}{2}$

$\sin\varphi=-\cos\langle\vec n,\vec s\rangle$

$\therefore \sin\varphi=|\cos\langle\vec n,\vec s\rangle|=\frac{|\vec n\cdot\vec s|}{|\vec n||\vec s|}$

(二) 距离

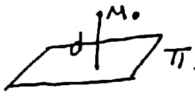
1. 两点之间距离

设 $A(x_1,y_1,z_1)$ $B(x_2,y_2,z_2)$ 则 A·B 之距

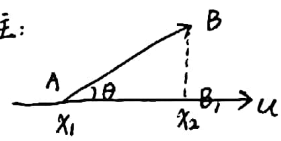
$d=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2+(z_2-z_1)^2}$

2. 点到平面距离

设 $\pi: Ax+By+Cz+D=0$ $M_0(x_0,y_0,z_0)\notin\pi$

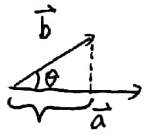


注:



$x_2-x_1=Prj_u\vec{AB}$

$=|\vec{AB}|\cos\theta$

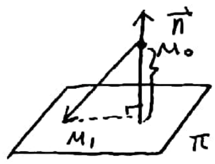


$Prj_{\vec a}\vec b=|\vec b|\cos\langle\vec a,\vec b\rangle$

$=\frac{|\vec a||\vec b|\cos\langle\vec a,\vec b\rangle}{|\vec a|}$

$=\frac{\vec a\cdot\vec b}{|\vec a|}$

即 $Prj_{\vec a}\vec b=\frac{\vec a\cdot\vec b}{|\vec a|}$



$\forall M_1(x_1,y_1,z_1)\in\pi$

$\vec{M_0M_1}=x_1-x_0, y_1-y_0, z_1-z_0$

$$Prj_{\vec n}\vec{M_0M_1}=\frac{\vec n\cdot\vec{M_0M_1}}{|\vec n|}=\frac{A(x_1-x_0)+B(y_1-y_0)+C(z_1-z_0)}{\sqrt{A^2+B^2+C^2}}$$

$$=\frac{(Ax_1+By_1+Cz_1)-(Ax_0+By_0+Cz_0)}{\sqrt{A^2+B^2+C^2}}$$

$\because M_1\in\pi$ $\therefore Ax_1+By_1+Cz_1+D=0$

$\therefore Ax_1+By_1+Cz_1=-D$

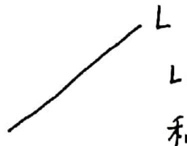
$$Prj_{\vec n}\vec{M_0M_1}=-\frac{Ax_0+By_0+Cz_0+D}{\sqrt{A^2+B^2+C^2}}$$

$$d=|Prj_{\vec{n}}\overrightarrow{M_0M_1}|=\frac{|Ax_0+By_0+Cz_0+D|}{\sqrt{A^2+B^2+C^2}}$$

例4：求点 $M_1(1,2,-1)$ 到平面 $\pi: x-y+z-4=0$ 的距离

$$d=\frac{|1-2-1-4|}{\sqrt{1^2+(-1)^2+1^2}}=\frac{6}{\sqrt{3}}=2\sqrt{3}.$$

(三) 平面束



L为直线，经过L的所有平面称为平面束.

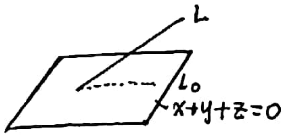
设 $L:\begin{cases}A_1x+B_1y+C_1z+D_1=0\\A_2x+B_2y+C_2z+D_2=0\end{cases}$

过L平面束为：$\pi: A_1x+B_1y+C_1z+D_1+\lambda(A_2x+B_2y+C_2z+D_2)=0$

即 $\pi: (A_1+A_2\lambda)x+(B_1+\lambda B_2)y+(C_1+\lambda C_2)z+D_1+\lambda D_2=0$

例5：求直线 $L:\begin{cases}x+y-z-1=0\\x-y+z+1=0\end{cases}$

在平面 $x+y+z=0$ 上的投影直线



~~解~~：

解：1° 过L的平面束为 $\pi: x+y-z-1+\lambda(x-y+z+1)=0$

即 $\pi: (\lambda+1)x+(1-\lambda)y+(\lambda-1)z+\lambda-1=0$

2° 由 $\{\lambda+1, 1-\lambda, \lambda-1\}\cdot\{1,1,1\}=0\Rightarrow\lambda=-1$

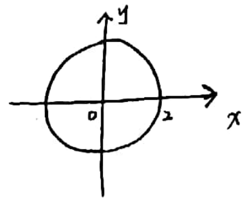
$\therefore$ 投影直线 $L_0:\begin{cases}2y-2z-2=0\\x+y+z=0\end{cases}$ 即 $L_0:\begin{cases}y-z-1=0\\x+y+z=0\end{cases}$

8.5 空间曲面及方程

$\Sigma: F(x,y,z)=0$

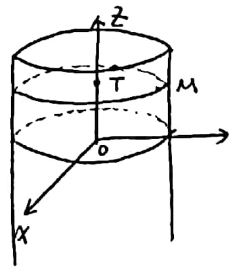
一、柱面. $x^2+y^2=4$

2-dim:



3-dim：求到z轴距离为2的点、形成的曲面.

$\forall M(x,y,z)\in\Sigma$ $T(0,0,z)$

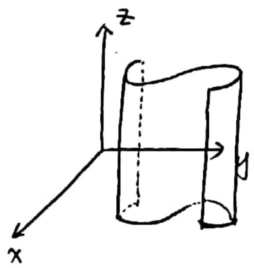


$|MT|=2$

$\Rightarrow\sqrt{x^2+y^2}=2$

$\Sigma: x^2+y^2=4$

1. $\Sigma: F(x,y)=0$ 为母线平行于z轴的柱面.



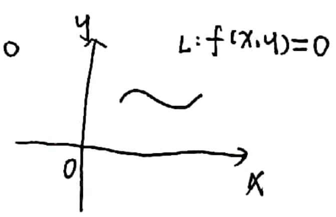
柱面 $\Sigma: F(x,y)=0$ 在 xoy 面内的投影曲线为

$L:\begin{cases}F(x,y)=0\\z=0\end{cases}$

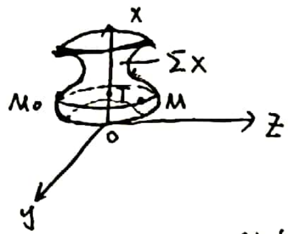
2. $\Sigma: G(y,z)=0$ 为母线平行于x轴的柱面

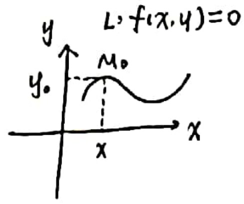
☆ 二、旋转曲面.

设 $L:\begin{cases}f(x,y)=0\\z=0\end{cases}$



① 设L绕 x轴旋转一周而成的曲面为Σx





$\forall M(x,y,z)\in\Sigma x$ $\cancel{M_0(x,y_0,z_0)}$

$M_0(x,y_0,0)\in L$ $T(x,0,0)$

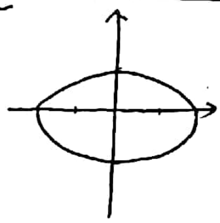
$|MT|=|M_0T| \Rightarrow \sqrt{y_0^2}=\sqrt{y^2+z^2}$

$\Rightarrow \cancel{y_0=\pm y_0+z}$ $y_0=\pm\sqrt{y^2+z^2}$

$\because M_0\in L$ $\therefore f(x,y_0)=0$

$\therefore \Sigma x = f(x,\pm\sqrt{y^2+z^2})=0$ $\boxed{\text{绕谁转谁不变}}$

例1 $L:\begin{cases}\dfrac{x^2}{4}+y^2=1\\ z=0\end{cases}$



求Σx

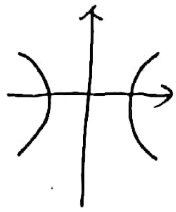
解：$\Sigma x: \dfrac{x^2}{4}+\left(\pm\sqrt{y^2+z^2}\right)^2=1$

即$\Sigma x: \dfrac{x^2}{4}+y^2+z^2=1$

② L $\begin{cases}f(x,y)=0\\ z=0\end{cases}$ 绕y轴旋转一周而成曲面Σy

$\Sigma y: f(\pm\sqrt{x^2+z^2},\ y)=0$

例2：$L:\begin{cases}\dfrac{x^2}{9}-\dfrac{y^2}{4}=1\\ z=0\end{cases}$



求Σx, Σy

解：$\Sigma x: \dfrac{x^2}{9}-\dfrac{(\pm\sqrt{y^2+z^2})^2}{4}=1 \Rightarrow \dfrac{x^2}{9}-\dfrac{y^2}{4}-\dfrac{z^2}{4}=1$

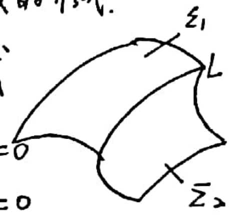
$\Sigma y: \dfrac{(\pm\sqrt{x^2+z^2})^2}{9}-\dfrac{y^2}{4}=1 \Rightarrow \dfrac{x^2+z^2}{9}-\dfrac{y^2}{4}=1$

8.6 空间曲线及其方程

一、空间曲线的形式

(一) 一般形式

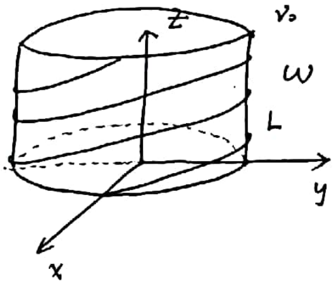
$L:\begin{cases}F(x,y,z)=0\\ G(x,y,z)=0\end{cases}$



(二). 参数式

如：L:

$\begin{cases}x=\cancel{R\omega t}R\cos\omega t\\ y=R\sin\omega t\\ z=V_0t\end{cases}$



$L:\begin{cases}x=\varphi(t)\\ y=\psi(t)\\ z=\omega(t)\end{cases}$ 一参数式

如：$L:\begin{cases}x^2+y^2=1\\ x+y-z-2=0\end{cases}$

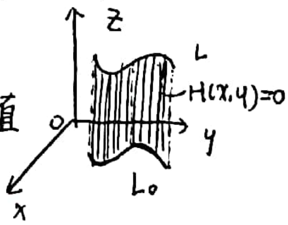
化为参数式

解：$L:\begin{cases}x=\cos t\\ y=\sin t\\ z=\sin t+\cos t-2\end{cases}$

二、曲线的特殊情形—直线、

(一) 点向式（对称式） (二)参数式.
(三) 一般式

三、投影曲线（L向xoy面铅直投影）

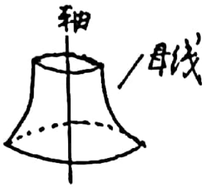


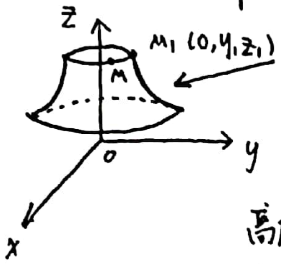
设 $L:\begin{cases}F(x,y,z)=0\\ G(x,y,z)=0\end{cases}$

由$\begin{cases}F(x,y,z)=0\\ G(x,y,z)=0\end{cases} \xrightarrow{\text{消}z} H(x,y)=0$

$L_0:\begin{cases}H(x,y)=0\\ z=0\end{cases}$

$\boxed{\text{旋转曲面}}$





$f(y,z)=0$

$M_1(0,y_1,z_1)$ $f(y_1,z_1)=0$

$M(x,y,z)$

高度相同. $z=z_1$.

圆上各点到圆心距离相等

$\Rightarrow y_1^2=x^2+y^2=d^2$

$\Rightarrow y_1=\pm\sqrt{x^2+y^2}$ 又: $z=z_1$

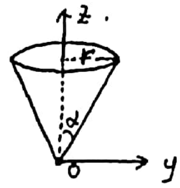
旋转曲面方程: $f(\pm\sqrt{x^2+y^2},z)=0$

$z_0=\frac{y}{\tan\alpha}=y_0\cot\alpha$

$x^2+y^2=r^2$ $r=\pm\sqrt{x^2+y^2}$

$z=\pm\sqrt{x^2+y^2}\cot\alpha$

$\Rightarrow z^2=\cot^2\alpha\,(x^2+y^2)$.



例: $\frac{x^2}{a^2}-\frac{z^2}{c^2}=1$

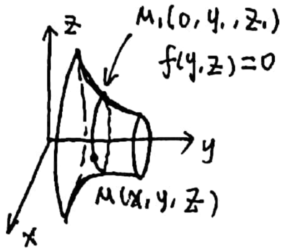
z轴: $\frac{x^2+y^2}{a^2}-\frac{z^2}{c^2}=1$.

x轴: $\frac{x^2}{a^2}-\frac{y^2+z^2}{c^2}=1$.

$y=y_1$ $|z_1|=\sqrt{x^2+z^2}$

$\therefore z_1=\pm\sqrt{x^2+z^2}$

$f(y,\pm\sqrt{x^2+z^2})=0$



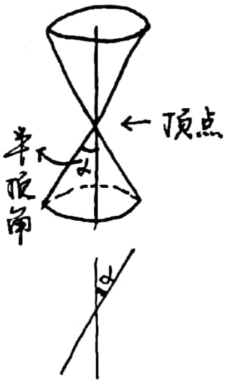
$\boxed{\text{绕谁转,谁不变}}$

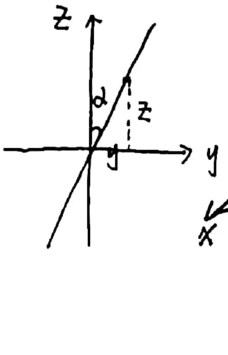
yoz平面 $f(y,z)=0$ <
- 绕z轴: $f(\pm\sqrt{x^2+y^2},z)=0$
- 绕y轴: $f(y,\pm\sqrt{x^2+z^2})=0$

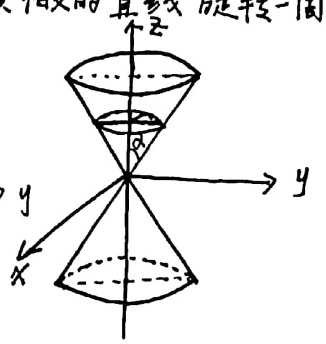
xoy平面 $f(x,y)=0$ <
- 绕x轴: $f(x,\pm\sqrt{y^2+z^2})=0$
- 绕y轴: $f(\pm\sqrt{x^2+z^2},y)=0$

xoz平面 $f(x,z)=0$ <
- 绕x轴: $f(x,\pm\sqrt{y^2+z^2})=0$
- 绕z轴: $f(\pm\sqrt{x^2+y^2},y)=0$

圆锥面: 一条直线,围绕与其相交的直线旋转一周.







$\boxed{\text{二次曲面}}$

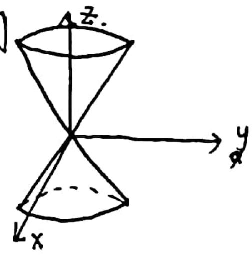
$F(x,y,z)=0$ 三元二次.

1) 椭圆~~曲面~~锥面 $\frac{x^2}{a^2}+\frac{y^2}{b^2}=z^2$.

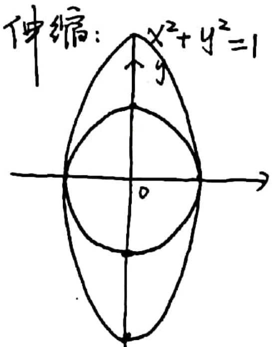
令 $z=t$ $\frac{x^2}{a^2t^2}+\frac{y^2}{b^2t^2}=1$

横截面为椭圆

(截痕法)



伸缩: $x^2+y^2=1$ $\longrightarrow$ 沿y轴伸缩2倍



$y\to\frac{1}{2}y$ $x^2+\frac{y^2}{4}=1$

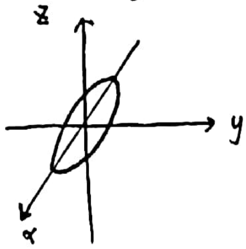
$x^2+y^2=1 \xrightarrow[\text{伸缩2倍}]{\text{沿x轴}} (2x)^2+y^2=1$

$\frac{x^2}{4}+\frac{y^2}{1}=1$

2) 椭球面 $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$

$\boxed{\text{法一}}$

xoz: $\frac{x^2}{a^2}+\frac{z^2}{c^2}=1$



绕z轴旋转

$$\frac{x^2+y^2}{a^2}+\frac{z^2}{c^2}=1$$

沿y轴伸缩 $\frac{b}{a}$ 倍

$$\frac{x^2}{a^2}+\frac{(\frac{a}{b}y)^2}{a^2}+\frac{z^2}{c^2}=1$$

$$\Rightarrow \frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$$

$\boxed{\text{法二}}$ $x^2+y^2+z^2=a^2$ 沿z轴伸缩 $\frac{c}{a}$ 倍

$$\Rightarrow x^2+y^2+\frac{a^2}{c^2}z^2=a^2 \Rightarrow \frac{x^2+y^2}{a^2}+\frac{z^2}{c^2}=1$$

沿y轴伸缩 $\frac{b}{a}$ 倍

$$\Rightarrow \frac{x^2}{a^2}+\frac{1}{a^2}\frac{a^2}{b^2}y^2+\frac{z^2}{c^2}=1$$

$$\therefore \frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$$

※: 伸缩多少倍，代入时取倒数

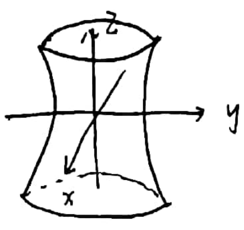
3) 单叶双曲面. $\frac{x^2}{a^2}+\frac{y^2}{b^2}-\frac{z^2}{c^2}=1$

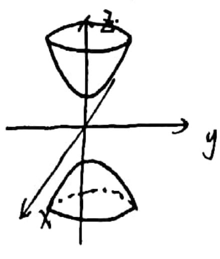
$\frac{x^2}{a^2}-\frac{z^2}{c^2}=1$ 沿z轴旋转

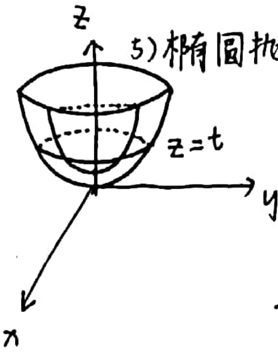
$\frac{x^2+y^2}{a^2}-\frac{z^2}{c^2}=1$ 沿y轴伸缩 $\frac{b}{a}$ 倍

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}-\frac{z^2}{c^2}=1$$

4) 双叶双曲面 $\frac{x^2}{a^2}+\frac{y^2}{b^2}-\frac{z^2}{c^2}=-1$







5) 椭圆抛物面. $\frac{x^2}{a^2}+\frac{y^2}{b^2}=z$.

~~z=t~~ x=1 $z=\frac{y^2}{b^2}+\frac{1}{a^2}$

椭圆锥面: $\frac{x^2}{a^2}+\frac{y^2}{b^2}=z^2$

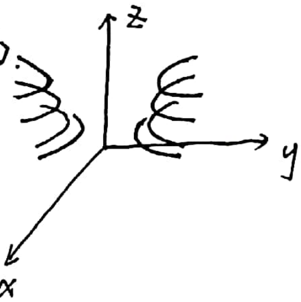
令 $y=0$ $\frac{x}{a}=\pm z$.

一次表达式 图像直线

而 $\frac{x^2}{a^2}+\frac{y^2}{b^2}=z$ 令 $y=0$ $z=\frac{x^2}{a^2}$

二次函数 → 抛物面

6) 双曲抛物面: $\frac{x^2}{a^2}-\frac{y^2}{b^2}=z$



~~x=t~~

$\boxed{\text{曲面及其方程}}$

例1: 球心 $M_0(x_0, y_0, z_0)$. 半径为R的球面

$|MM_0|=R$ $(x-x_0)^2+(y-y_0)^2+(z-z_0)^2=R^2$

球心在原点: $x^2+y^2+z^2=R^2$

例2: $x^2+y^2+z^2-2x+4y=0$

$(x-1)^2+(y+2)^2+z^2=5$

球心 $(1,-2,0)$, $R=\sqrt{5}$

$$Ax^2+Ay^2+Az^2+Dx+Ey+Fz+G=0 \quad (A\neq 0)$$

$$x^2+y^2+z^2+(\frac{D}{A})x+(\frac{E}{A})y+(\frac{F}{A})z=-\frac{G}{A}$$

$$(x+\frac{D}{2A})^2+(y+\frac{E}{2A})^2+(z+\frac{F}{2A})^2=-\frac{G}{A}+\frac{D^2+E^2+F^2}{4A^2}$$

$\frac{D^2+E^2+F^2-4AG}{4A^2}>0$ ∴ $D^2+E^2+F^2>4AG$.